

Integrations & platform capabilities: what's real, what's a stub, what's missing

Every claim below is backed by a direct code read — file and line, or an explicit "no matches found." Two passes: 20 third-party integration categories, and 20 core capabilities expected of a Salesforce/HubSpot/Zoho-tier CRM. 40 categories total.

40

CATEGORIES CHECKED

7

FULLY EXISTS

11

PARTIAL / STUBBED

22

ABSENT

Headline. Communications infrastructure is genuinely strong — SMTP, SMS (7 providers), WhatsApp, RCS with template approval, and telephony (EasyGoIVR with click-to-call) are all fully wired, not stubs. Payments and lead-source integrations are real but incomplete (Razorpay works, PayU is UI-only; Meta/Instagram work, LinkedIn is UI-only with no backend route). Everything else that makes a CRM feel "enterprise-grade" — analytics/attribution, e-signature, video conferencing, a general-purpose developer API, 2FA, GDPR tooling, a workflow-rule builder, custom fields, audit trails on records, forecasting — is absent. The product is a strong operational lead-to-admission engine with a comms stack most CRMs charge extra for; it is not yet a platform in the Salesforce/HubSpot sense.

PART 1

Integrations

#	CATEGORY	STATUS	EVIDENCE
1	Payment gateways	● PARTIAL	Razorpay live (index.js:5795 creates real payment links). PayU is UI-only — settings fields exist, no API calls found. No Stripe.
2	Email (SMTP)	● EXISTS	Full mailer, used for alerts/OTP/reports/receipts, dedicated test endpoint (index.js:413).
3	SMS gateway	● EXISTS	7 real providers wired (msg91/twilio/plivo/textlocal/gupshup/kaleyra/kaleya) — index.js:4877 .

#	CATEGORY	STATUS	EVIDENCE
4	WhatsApp	● EXISTS	Meta Cloud API webhook + outbound sending (index.js:4715). Not a BSP-specific integration but functional.
5	Voice / telephony	● EXISTS	EasyGoIVR — click-to-call, token refresh, call-stat webhook (index.js:4299–4440). No other voice provider.
6	Social/ads lead sources	● PARTIAL	Meta + Instagram fully wired. Google Ads: webhook ingest only, OAuth fields never read. LinkedIn: settings UI + advertised webhook URL, no matching backend route exists .
7	Analytics / tracking	● ABSENT	GA4 fields exist in Integrations UI but zero backend usage. No Meta Pixel/CAPI. No UTM parameter capture anywhere.
8	Cloud storage / backup	● EXISTS	AWS S3, real <code>S3Client</code> calls, daily 3am cron (index.js:7169–7263).
9	Spreadsheet sync	● EXISTS	Google Sheets → leads, one-way, cron-scheduled (index.js:6432).
10	AI / LLM	● EXISTS	Anthropic Claude powers Ask CU AI — real, but no settings card in the Integrations UI; key must be set via an undocumented path.
11	Calendar sync	● PARTIAL	Internal <code>events</code> table only. No Google Calendar/Outlook/iCal OAuth sync, no .ics export.
12	Video conferencing	● ABSENT	No Zoom/Meet/Teams integration anywhere.
13	E-signature	● ABSENT	No DocuSign or equivalent. Document routes are upload/verify only.
14	SSO / identity	● PARTIAL	Consumer Google OAuth login exists. No SAML, Okta, or Azure AD.
15	Public API for 3rd parties	● PARTIAL	Two narrow shared-secret endpoints (lead lookup, payment status) with a hardcoded fallback key in source — not a general API-key-management system.

#	CATEGORY	STATUS	EVIDENCE
16	Zapier / iPaaS	• ABSENT	No Zapier, Make, or generic iPaaS connector.
17	Accounting / ERP	• ABSENT	No Tally/QuickBooks/SAP/Zoho Books. "ERP access in the post-admission flow is a status flag + email explicitly documented as not a real integration.
18	Template approval workflows	• PARTIAL	RCS has full template-approval + DLR webhooks. No Meta WhatsApp official template CRUD.
19	Credit / KYC / ID verification	• ABSENT	No CIBIL, Aadhaar, or PAN verification API. KYC fields are stored, never verified.
20	LMS / student portal	• ABSENT	No external LMS (Moodle/Canvas) connection. Explicitly acknowledged in code comments as a future roadmap item.

PART 2

Core platform capabilities

#	CATEGORY	STATUS	EVIDENCE
1	RBAC granularity	• PARTIAL	Custom roles/teams can be created, but permission checks are hardcoded role-name string comparisons — a custom role is just a label with no enforced permissions.
2	Audit trail on records	• PARTIAL	Platform-admin cross-tenant actions are logged. No field-level change history on a lead/application record itself.
3	Custom fields	• ABSENT	Zero matches for a custom-field mechanism. Every field is a hardcoded DB column.
4	Workflow / automation builder	• ABSENT	No generic "when X happens, do Y" rule engine. Only the hardcoded Drip Workflows campaign feature.

#	CATEGORY	STATUS	EVIDENCE
5	Dedup / merge tool	• ABSENT	Create-time duplicate detection exists; no tool to find/merge two already-existing duplicate leads.
6	Bulk import / export	• PARTIAL	Leads import well (with column mapping). No import/export routes for users or payments. No scheduled/recurring export.
7	Two-factor authentication	• ABSENT	No TOTP/2FA anywhere — password or Google OAuth only.
8	GDPR / privacy tooling	• ABSENT	No consent tracking, data-subject export, or right-to-be-forgotten deletion implemented (mentioned only in docs, not code).
9	Backup / restore	• PARTIAL	S3 backup is real and admin-triggerable. No restore capability exists — recovery requires manual SSH + <code>pg_restore</code> .
10	Custom report builder	• ABSENT	All reports are fixed pages; no field/filter/chart-type builder.
11	General developer API	• ABSENT	No API-key issuance, scoping, or documentation for external developers — distinct from the narrow endpoints noted in Part 1 #15.
12	Mobile app / PWA	• ABSENT	No React Native, Capacitor, service worker, or PWA manifest.
13	SLA tracking / escalation	• ABSENT	Only the fixed "morning untouched leads" cron — no configurable SLA-hours or escalation rules.
14	Forecasting / territory mgmt	• ABSENT	No weighted-pipeline revenue forecasting or territory-assignment logic beyond round-robin.
15	Email open/click tracking	• ABSENT	Delivery status only — no tracking pixel or click-through logging.

#	CATEGORY	STATUS	EVIDENCE
16	Scheduling links	• ABSENT	No Calendly-style self-service booking page for students.
17	Rate limiting	• PARTIAL	Exists, but scoped only to <code>/api/auth</code> and <code>/api/webhooks</code> — not the public inquiry form or document upload.
18	Multi-language / i18n	• ABSENT	No translation framework; all UI text is hardcoded English.
19	Sandbox / staging mode	• ABSENT	No environment flag for testing workflows without touching real data.
20	Data retention / archival	• ABSENT	No archiving of stale leads — everything lives forever in the primary table (only a destructive full "Production Reset" exists).

3 Where to spend effort first

Grouped by real-world payoff versus build cost, not by category order above.

● Quick wins — finish what's already 80% built

- Give Ask CU AI (Anthropic) its own Integrations settings card — it's fully working but invisible to any Admin who doesn't already know the hidden config path
- Add a real `/api/webhooks/linkedin-leads` route — the UI and webhook URL are already advertised, the backend just isn't there
- Wire the PayU fields that already exist in the UI, or remove them if PayU isn't actually planned
- Add restore documentation/tooling on top of the existing S3 backups — the hard part (getting data out safely) is already solved

● Medium build — real gaps a growing team will hit soon

- UTM capture + Meta Pixel/CAPI + GA4 wiring — you're already paying for ads; right now there's no way to measure cost-per-lead or cost-per-admission
- Field-level audit trail on leads/applications — "who changed this and when" is one of the first things a compliance or ops review asks for

- A merge-duplicates tool — create-time dedup is good, but duplicates from bulk imports/API sources will still slip through and need cleanup
- Two-factor authentication — cheap to add (TOTP), meaningfully raises the bar against account takeover on Admin accounts

● **Major initiatives — only worth it once the above is done**

- A generic workflow/automation rule engine ("when X happens, do Y") — this is what would make the "digital ecosystem, not a database" pitch actually true across every module, not just payments
- Custom fields + a report builder — real platform-tier features, but a significant schema/UI investment
- A documented public API with proper key management — only worth it once there's an actual external system wanting to integrate
- Enterprise SSO (SAML/Okta) — matters once you're selling into institutions with their own IT policy, not before

Prepared from a full-repo audit of the CCRM codebase — every verdict above is backed by a specific file, line, or an explicit "no matches found."